Ratios, Rates & Data

ANSWER KEY

Section 1: Ratios & Proportional Reasoning

1 The ratio of red, blue, and yellow marbles in a bag is $3:4:5$. If there are 48 marbles in total, how many blue marbles are there?

a. 12

b. 16 ✓

c. 20

d. 24

Total ratio parts $= 3+4+5 = 12$. Each part $= \frac{48}{12} = 4$. Blue marbles $= 4 \times 4 = 16$.

2 If 8 workers can assemble 120 computer units in 5 hours, how many units can 12 workers assemble in 6 hours at the same rate?

a. 180

b. 216 ✓

c. 240

d. 288

Rate per worker $= \frac{120}{8 \times 5} = \frac{120}{40} = 3$ units per hour. For 12 workers working 6 hours: $12 \times 6 \times 3 = 216$ units.

3 A map has a scale of 1 cm $: 25$ km. If two cities are 7.2 cm apart on the map, what is the actual distance between them?

a. 150 km

b. 175 km

c. 180 **km** ✓

d. 200 km

Actual distance $= 7.2 \times 25 = 180$ km.

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Section 2: Percent Changes & Financial Calculations

4 An item originally priced at 80 is on sale for 25% off. An additional 10% tax is then added to the sale price. What is the final total?

a. 64

b. 66 ✓

c. 68

d. 72

Sale price $= 80 \times 0.75 = 60$. Adding 10% tax: $60 \times 1.10 = 66$.

5 A company's revenue increased by 20% in year 1 and then decreased by 15% in year 2. What was the net percent change over the two-year period?

a. $+2\%$ ✓

b. $+5\%$

c. -2%

d. $+3.5\%$

Multiplier $= 1.20 \times 0.85 = 1.02$. This corresponds to a net increase of 2%.

6 If 40% of x is equal to 30% of y, what is the ratio $\frac{x}{y}$?

a. $\frac{3}{4}$ ✓

b. $\frac{4}{3}$

c. $\frac{3}{7}$

d. $\frac{7}{3}$

$0.40x = 0.30y \Rightarrow \frac{x}{y} = \frac{0.30}{0.40} = \frac{3}{4}$.

Section 3: Speed, Distance & Average Rates

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7 A driver travels from City A to City B at an average speed of 60 km/h and returns along the same route at 40 km/h. What is the average speed for the round trip?

a. 48 **km/h** ✓

b. 50 km/h

c. 52 km/h

d. 54 km/h

Harmonic mean of speeds: $\frac{2v_1v_2}{v_1+v_2} = \frac{2(60)(40)}{60+40} = \frac{4800}{100} = 48$ km/h.

8 Train A leaves a station traveling east at 80 km/h. Two hours later, Train B leaves the same station traveling east on a parallel track at 120 km/h. How many hours after Train B departs will it overtake Train A?

a. 3

b. 4 ✓

c. 5

d. 6

Head start of Train A $= 80 \times 2 = 160$ km. Relative closing speed $= 120 - 80 = 40$ km/h. Time $= \frac{160}{40} = 4$ hours.

9 A cyclist rides 15 km at 20 km/h and then another 25 km at 25 km/h. What is the total time for the trip in minutes?

a. 90 min

b. 105 **min** ✓

c. 120 min

d. 135 min

Time $1 = \frac{15}{20} = 0.75$ h $= 45$ min. Time $2 = \frac{25}{25} = 1.0$ h $= 60$ min. Total time $= 45 + 60 = 105$ minutes.

Section 4: Data Tables & Two-Way Frequency

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10 In a survey of 200 high school seniors, 120 plan to attend college, 50 plan to work, and 30 are undecided. If a student is chosen at random, what is the probability that the student does NOT plan to work?

a. $\frac{1}{4}$

b. $\frac{3}{4}$ ✓

c. $\frac{3}{5}$

d. $\frac{17}{20}$

Students not planning to work $= 200 - 50 = 150$. Probability $= \frac{150}{200} = \frac{3}{4}$.

11 A two-way table shows that of 80 male students, 32 play a sport, and of 120 female students, 48 play a sport. What proportion of all students play a sport?

a. 0.35

b. 0.40 ✓

c. 0.45

d. 0.50

Total sports players $= 32 + 48 = 80$. Total students $= 80 + 120 = 200$. Proportion $= \frac{80}{200} = 0.40$.

12 Given that a student plays a sport from the above group, what is the conditional probability that the student is female?

a. 0.40

b. 0.50

c. 0.60 ✓

d. 0.65

Among the 80 sports players, 48 are female: $\frac{48}{80} = \frac{6}{10} = 0.60$.

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